

# Neurexins control the strength and precise timing of glycinergic inhibition in the auditory brainstem

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## Abstract

Neurexins play diverse functions as presynaptic organizers in various glutamatergic and GABAergic synapses. However, it remains unknown whether and how neurexins are involved in shaping functional properties of the glycinergic synapses, which mediate prominent inhibition in the brainstem and spinal cord. To address these issues, we examined the role of neurexins in a model glycinergic synapse between the principal neuron in the medial nucleus of the trapezoid body (MNTB) and the principal neuron in the lateral superior olive (LSO) in the auditory brainstem. Combining RNAscope with stereotactic injection of AAV-Cre in the MNTB of neurexin1/2/3 conditional triple knockout mice, we showed that MNTB neurons highly express all isoforms of neurexins although their expression levels vary remarkably. Selective ablation of all neurexins in MNTB neurons not only reduced the amplitude but also altered the kinetics of the glycinergic synaptic transmission at LSO neurons. The synaptic dysfunctions primarily resulted from an impaired  $\text{Ca}^{2+}$  sensitivity of release and a loosened coupling between voltage-gated  $\text{Ca}^{2+}$  channels and synaptic vesicles. Together, our current findings demonstrate that neurexins are essential in controlling the strength and temporal precision of the glycinergic synapse, which therefore corroborates the role of neurexins as key presynaptic organizers in all major types of fast chemical synapses.

## eLife assessment

This study provides **important** insights into the role of neurexins as regulators of synaptic strength and timing at the glycinergic synapse between neurons of the medial nucleus of the trapezoid body and the lateral superior olive, key components of the auditory brainstem circuit involved in computing sound source location from differences in the intensity of sounds arriving at the two ears. Through an elegant combination of genetic manipulation, fluorescence in-situ hybridization, ex vivo slice electrophysiology, pharmacology and optogenetics, the authors provide **compelling** and rigorous evidence to support their claims. While further work is needed to reveal the mechanistic basis by which neurexins influence glycinergic neurotransmission, this work will be of interest to both auditory and synaptic neuroscientists.

## Introduction

Neurexins are evolutionarily conserved synaptic adhesion molecules and their genetic mutations are highly associated with autism and schizophrenia (Gomez et al., 2021 [↗](#); Sudhof, 2008 [↗](#); 2017 [↗](#)). In the vertebrates, neurexins are encoded by three genes *Nrxn1*, *Nrxn2*, *Nrxn3*, each of which contains distinct promoters for expressing longer  $\alpha$ - and shorter  $\beta$ -neurexins (Ullrich et al., 1995 [↗](#); Ushkaryov et al., 1992 [↗](#)). Both  $\alpha$ - and  $\beta$ -neurexins are critical and non-redundant in regulating the formation and function of synapses (Anderson et al., 2015 [↗](#); Missler et al., 2003 [↗](#); Sudhof, 2017 [↗](#)). Triple knockout (KO) of  $\alpha$ -neurexins causes a significant decrease in the density of GABAergic but not glutamatergic synapses while producing a strong reduction in both inhibitory and excitatory synaptic transmission (Missler et al., 2003 [↗](#)). Triple KO of  $\beta$ -neurexins reveals a significant phenotype only in glutamatergic synapses, but not in GABAergic synapses (Anderson et al., 2015 [↗](#)). Deletion of all neurexins, including *Nrxn1/2/3*  $\alpha$ - and  $\beta$ -neurexins, leads to profound but strikingly distinct phenotypes in different synapses, which further points to the diverse actions of neurexins in specific neurons (Chen et al., 2017 [↗](#); Luo et al., 2020 [↗](#)). Consistently, transcriptomic analysis has revealed that the expression profiles of various neurexins differ remarkably in excitatory neurons versus inhibitory neurons even among distinct subtypes of interneurons (Foldy et al., 2016 [↗](#); Fuccillo et al., 2015 [↗](#); Lukacsovich et al., 2019 [↗](#); Uchigashima et al., 2019 [↗](#)). Despite the multifaceted roles of neurexins widely observed in various glutamatergic and GABAergic synapses (Boxer and Aoto, 2022 [↗](#); Chen et al., 2017 [↗](#); Luo et al., 2021 [↗](#); Sudhof, 2017 [↗](#)), however, it remains largely unknown how neurexins may regulate the functional properties of the glycinergic synapses, which mediate essential inhibition in the spinal cord, brainstem and retina (Assareh et al., 2023 [↗](#); Brill et al., 2020 [↗](#); Chang et al., 2023 [↗](#); Feng et al., 1998 [↗](#); Foster et al., 2015 [↗](#); Gomeza et al., 2003a [↗](#); Gomeza et al., 2003b [↗](#); Zeilhofer et al., 2012 [↗](#)).

In the mature mammalian auditory brainstem, it is well-characterized that the principal neurons in the medial nucleus of the trapezoid body (MNTB) project glycinergic synaptic inputs to the lateral superior olive (LSO) (Fischer et al., 2019 [↗](#); Kotak et al., 1998 [↗](#); Nabekura et al., 2004 [↗](#); Sanes and Friauf, 2000 [↗](#)). These synapses are robust, precise and extremely reliable even during high frequency stimulation (Krachan et al., 2017 [↗](#)). Strong evidence has shown that the MNTB-LSO glycinergic synapses are subject to extensive refinement during development, including transmitter type switch, the elaboration of dendrites and axonal arbors followed by synaptic pruning, to achieve exquisite control of fast inhibition (Brill et al., 2020 [↗](#); Kim and Kandler, 2003 [↗](#); Kotak et al., 1998 [↗](#); Nabekura et al., 2004 [↗](#); Sanes and Friauf, 2000 [↗](#)). However, the molecular mechanisms of shaping their functional characteristic remain incompletely understood.

We utilized the triple *Nrxn1/2/3* conditional KO mice (*Nrxn123* cTKO) (Chen et al., 2017 [↗](#)) to ablate all neurexins in MNTB principal neurons with stereotactic injection of adeno-associated-virus (AAV) expressing Cre-recombinase. Combining this with the RNAscope technique, we verified that MNTB neurons highly express various isoforms of neurexins. Selective injection of AAV-Cre, but not AAV-EGFP, successfully blocked expression of all neurexins in MNTB neurons. Functionally, pan-neurexin deletion significantly impaired both the amplitude and the kinetics of the glycinergic synaptic transmission. Surprisingly, deletion of all neurexins led to a significantly higher number of glycinergic synapses innervating the LSO and higher frequency of spontaneous neurotransmitter release. Application of low extracellular  $\text{Ca}^{2+}$  blocked the glycinergic neurotransmission of both control and neurexin-deficient synapses, but with a significant difference in its effectiveness. Moreover, treatment with EGTA, a high-affinity  $\text{Ca}^{2+}$  chelator with slow binding kinetics, caused a much stronger inhibition of neurotransmission in neurexin-deficient synapses, suggesting a significant disruption of the tight coupling between voltage-gated  $\text{Ca}^{2+}$  channels and synaptic vesicles in the MNTB-LSO glycinergic synapses.

## Results

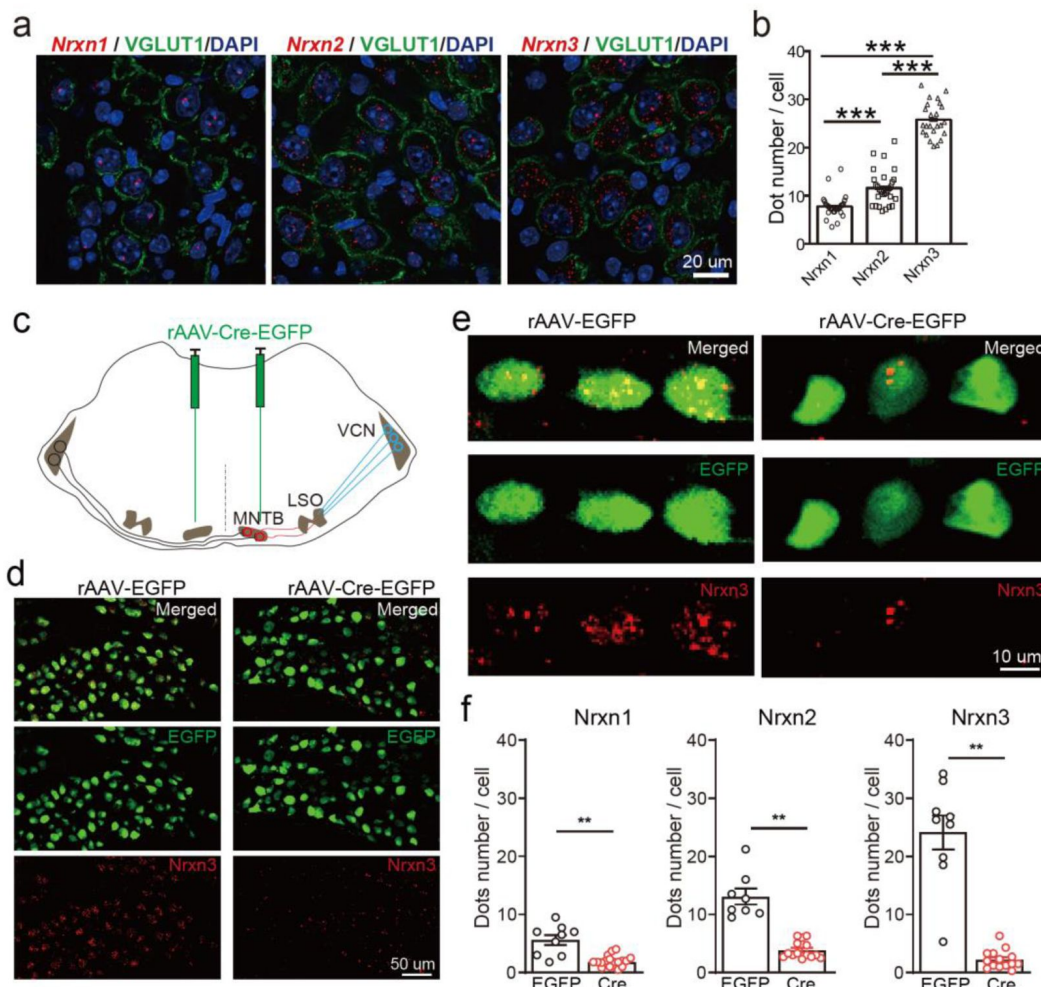
### Neurexins are highly expressed in MNTB neurons

To study the role and mechanism of neurexins in the glycinergic synapse between MNTB neurons and LSO neurons, which are critical in computing sound localization in mammalian auditory brainstem (Grothe et al., 2010 [↗](#)), we first characterized the expression of Nrnx1-3 in the principal neurons of the MNTB by fluorescent *in-situ* hybridization (FISH) using the RNAscope technique. As shown in **Figure 1** [↗](#), MNTB neurons in wild-type (WT) mice expressed Nrnx1, Nrnx2, and Nrnx3, although the expression level of each gene appeared to vary remarkably. As an initial exploration, we took advantage of Nrnx123 cTKO mice that enable deletion of all neurexins following the expression of Cre recombinase (Chen et al., 2017 [↗](#)), to bypass the functional redundancy of different neurexins. Neuronal specific deletion of all neurexins was achieved by stereotactic injection of AAV-Cre-EGFP into the MNTB at P0 (TKO). As a control, AAV-EGFP was injected (Ctrl). Successful virus injection and expression of Cre recombinase in MNTB neurons were confirmed by strong EGFP fluorescence (**Figure 1d-e** [↗](#)). FISH data further demonstrated that the recombinase dramatically blocked the expression of all neurexins, as compared to the normal expression of neurexins when EGFP was expressed (**Figure 1f** [↗](#), compared to 1b). Altogether, these data show that the glycinergic neurons in the MNTB highly express various neurexin genes, which can be specifically deleted by stereotactic injection of AAV-Cre in Nrnx123 cTKO mice.

### Neurexins are essential for dictating the strength and kinetics of the MNTB-LSO glycinergic synapse

Since neurexins play an essential role in orchestrating functional organization of various synapses including both glutamatergic and GABAergic synapses, we hypothesize that deletion of all neurexins may have a significant impact on glycinergic synapses. To test the hypothesis, we sought to examine how neurexins may regulate the function of the MNTB-LSO glycinergic synapse in the auditory brainstem. Because LSO neurons receive excitatory glutamatergic inputs and inhibitory GABAergic/glycinergic inputs, we used a pharmacological approach to isolate the glycinergic inputs. By blocking glutamatergic inputs with CNQX and AP5, we were able to selectively record inhibitory postsynaptic currents (IPSCs) from LSO neurons, which can be completely inhibited by strychnine, a potent glycine receptor inhibitor (**Figure S1** [↗](#)).

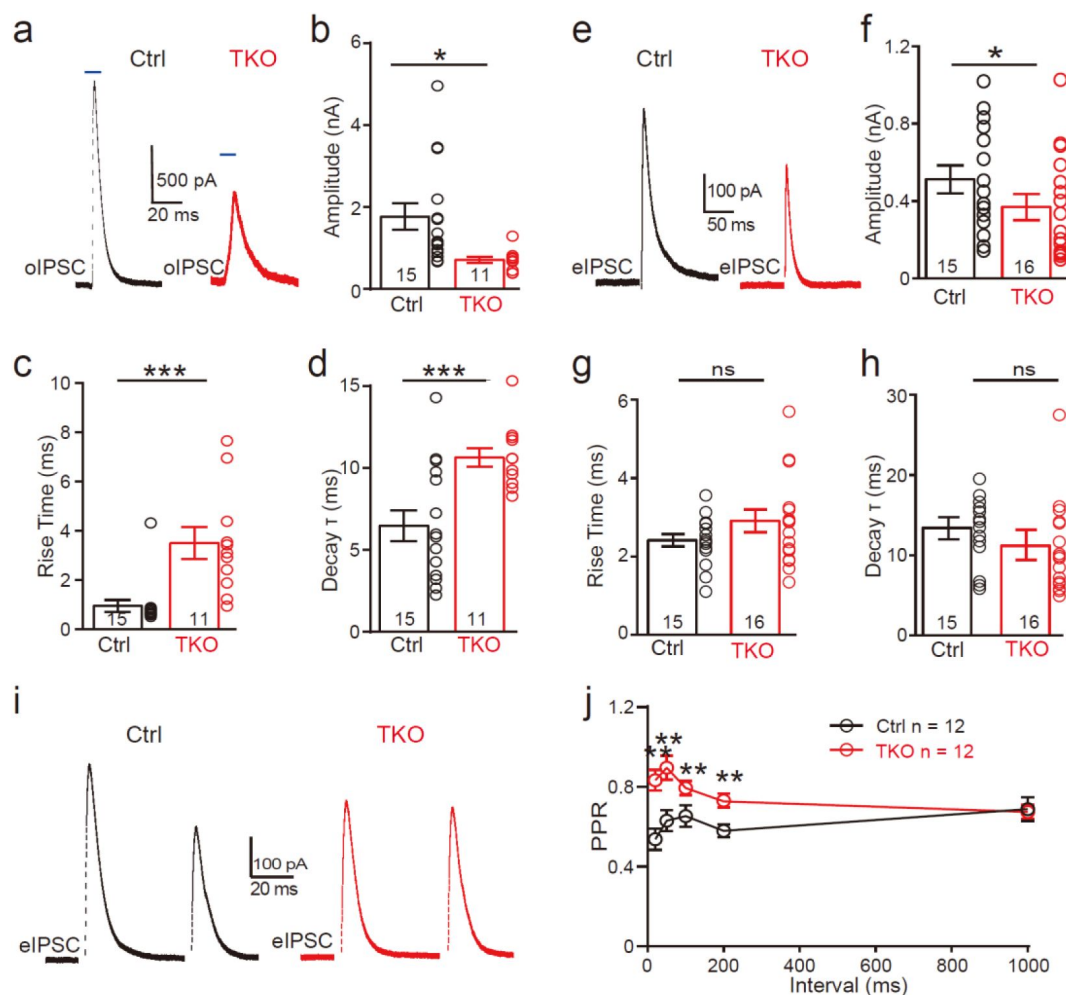
To specifically examine the role of neurexins in regulating the glycinergic synapse, we used an optogenetic approach in which two AAVs, AAV-Cre-EGFP and AAV-DIO-ChR2-mCherry, were co-injected in Nrnx123 cTKO mice. As a control, AAV-ChR2-EYFP was used for injection. Virus injection was performed at P0 and acute slices derived from these mice were analyzed at P13-14. We verified the successful expression of ChR2 by examining the fluorescence and further by recording optogenetically-evoked action potentials from the soma of MNTB neurons. Reliable action potential (AP) firing can be induced by single or repetitive high-frequency light stimulation (**Figure S1c** [↗](#)). In control mice, optogenetic stimulation of MNTB neurons robustly triggered large IPSCs recorded from LSO neurons (**Figure 2a-b** [↗](#)), which can be fully blocked by strychnine (**Figure S1d** [↗](#)) and are thus purely glycinergic. In TKO mice with co-injection of Cre and ChR2, however, optogenetic stimulation induced significantly smaller IPSCs (**Figure 2a-b** [↗](#)). On average, we observed ~60% reduction in the peak amplitude (**Figure 2b** [↗](#)), similar to previous findings at other different synapses (Chen et al., 2017 [↗](#); Luo et al., 2020 [↗](#)). Furthermore, the kinetics of IPSCs were dramatically reduced as indicated by significantly slower rise time and decay time in TKO mice as compared to the control (**Figure 2c-d** [↗](#)). Together, these data strongly suggest that neurexins are essential for the intact function of the glycinergic synapse.



**Fig. 1**

### Neurexins are highly expressed in MNTB neurons.

(a) Immunohistochemistry and RNAscope FISH showing expression of various isoforms of neurexins in the MNTB, which make glycinergic synapses with LSO neurons (b) Summary of expression of various isoforms of neurexins in the MNTB of WT mice. *Nrnx1* (n = 29), *Nrnx2* (n = 29), *P* = 0.0001; *Nrnx1* (n = 29), *Nrnx3* (n = 24), *P* = 0.0001; *Nrnx2* (n = 29), *Nrnx3* (n = 24), *P* = 0.0001. (c) Diagram of auditory brainstem and virus injection in *Nrnx123* cTKO mice. (d) Representative images of *Nrnx3* FISH in the MNTB of mice injected with AAV-EGFP or AAV-Cre-EGFP. Scale bar: 50  $\mu$ m. (e) Selected ROI in high magnification showing *Nrnx3* FISH in MNTB neurons expressing EGFP or Cre-EGFP. Scale bar: 10  $\mu$ m. (f) Summary of expression of various neurexins in the MNTB of *Nrnx123* cTKO mice injected with AAV-EGFP and AAV-Cre-EGFP. *Nrnx1*: EGFP (n = 9), Cre (n = 19), *P* = 0.0053; *Nrnx2*: EGFP (n = 8), Cre (n = 14), *P* = 0.0038; *Nrnx3*: EGFP (n = 9), Cre (n = 14), *P* = 0.0016. Data are means  $\pm$  SEM. Number of sections analyzed are indicated in the bars (b, f); Statistical differences were assessed by Student's *t* test. (\*\**P* < 0.01, \*\*\**P* < 0.001). Source data are provided as a Source Data file.



**Figure 2**

### Deletion of neuroligins reduces the strength and kinetics of glycinergic synapse.

(a) Representative IPSC traces of LSO neurons evoked by optogenetic stimulation of ChR2 expressed in MNTB neurons. (b-d) Summary of IPSC amplitude, rise time, and decay time constant. Ctrl (n = 15), TKO (n = 11),  $P = 0.0119$ ,  $P = 0.0004$ ,  $P = 0.002$ . unpaired two-sided t-test. (e-h) Same as a-d, except that IPSCs were evoked by afferent fiber stimulation. Ctrl (n = 15), TKO (n = 16),  $P = 0.0267$ ,  $P = 0.152$ ,  $P = 0.241$ . (i) Representative IPSCs traces of LSO neurons evoked by a pair of fiber stimulation at 50 ms interval. (j) Summary of paired-pulse ratio (PPR) of IPSCs in relationship to different paired pulse intervals. Ctrl (n = 12), TKO (n = 12),  $P < 0.0001$ , two-way ANOVA.

Data are means  $\pm$  SEM. Number of neuron are indicated in the bars (b-d, f-h) or graph (j). Statistical differences were assessed by Student's t test or two-way ANOVA test. (\* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ ).



To further address the physiological role of neurexins in action potential evoked neurotransmission at the glycinergic synapse, we applied electric stimulation to activate presynaptic fibers of MNTB neurons by placing a stimulating electrode in the midway between the MNTB and the LSO. Consistent with our optogenetic approach, afferent-evoked large IPSCs were predominantly glycinergic as they can also be fully blocked by strychnine (**Figure S1**). A modest but significant decrease in synaptic strength was observed in Nrnx123 TKO mice as compared to control mice (**Figure 2e-f**), confirming the important role of neurexins in the glycinergic synapses. The modest effect of the pan-neurexin deletion on synaptic strength in the MNTB-LSO synapses is likely due to the incomplete AAV virus transfection in MNTN neurons. Therefore, it is likely that a small fraction of MNTB neurons may still express relatively normal expression of neurexins. It is worth pointing out that extensive efforts were made to optimize our virus injection protocols by adjusting the injection parameters including the needle angle, virus titers, and volume. On average, it was estimated that approximately ~80% neurons in the MNTB expressed GFP and thus were infected by AAV-Cre. It is also possible that other sources of glycinergic inputs other than the MNTB afferents are likely to present but are obscured in wild-type animals by the predominant glycinergic inputs from MNTB neurons (Jalabi et al., 2013).

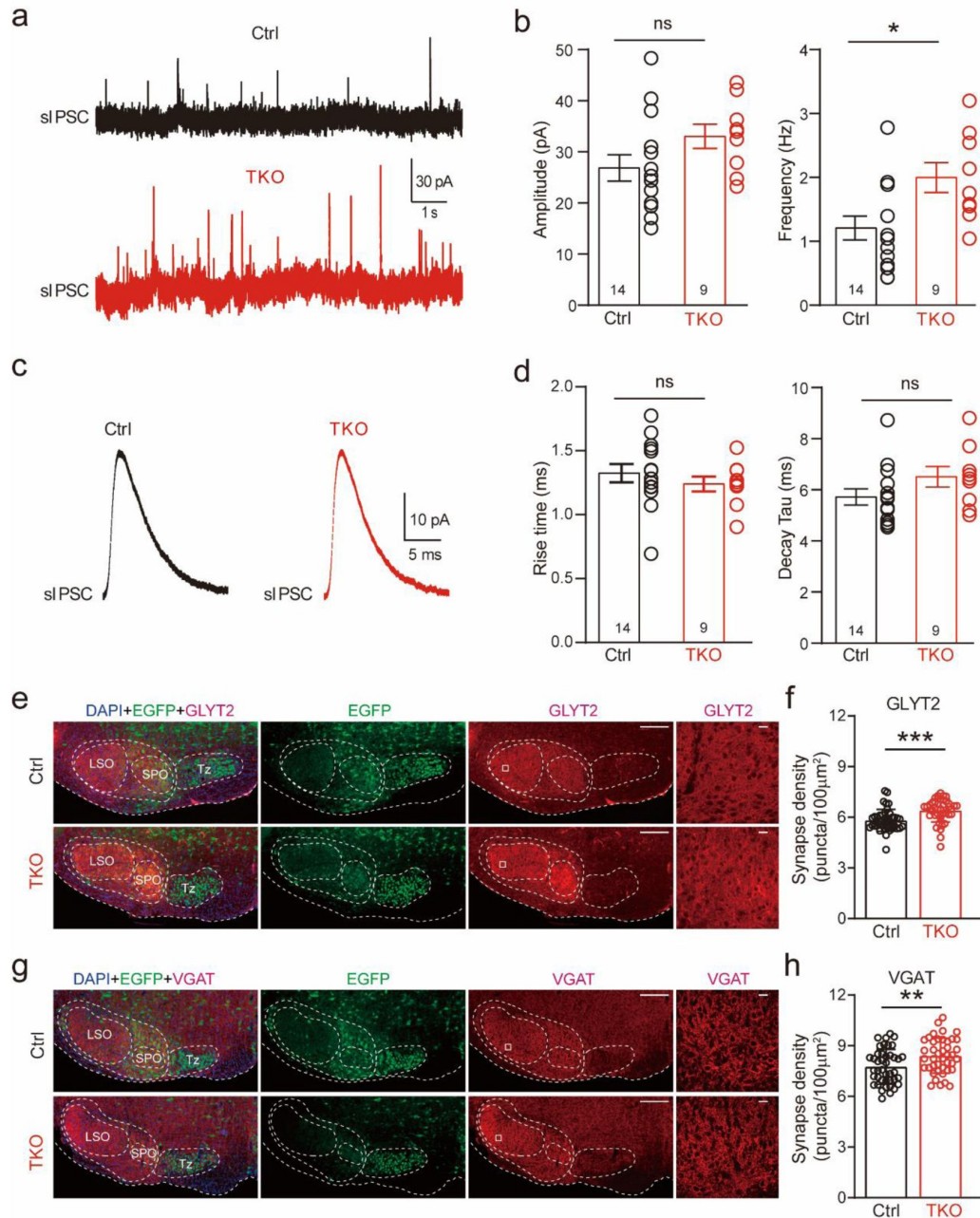
## The mechanism of neurexins for regulating the MNTB-LSO glycinergic synapse

Since neurexins play multifaceted roles in regulating pre- and/or post-synaptic functions, we explored the potential mechanisms by initially examining the paired-pulse ratio (PPR) of IPSCs, which has been routinely used for identifying a presynaptic origin of regulation. As shown in **Figure 2i**, IPSCs evoked by two consecutive stimuli at various intervals showed a paired-pulse depression in both control and TKO mice. However, the PPRs were significantly increased by deletion of neurexins (**Figure 2j**, **s2**), suggesting that neurexins are required for maintaining a high release probability of the glycinergic nerve terminal.

We then recorded and analyzed spontaneous IPSCs from LSO neurons. Compared to the control, there was no significant difference in both the amplitude and kinetics of sIPSCs (**Figure 3a-d**) in TKO mice, indicating the normal content of synaptic vesicle as well as the normal function of postsynaptic glycine receptors. Surprisingly, however, we found that deletion of all neurexins caused a small but significant increase in the frequency of sIPSCs (**Figure 3a-b**), which is opposite to previous reports of either no change or decrease in the miniature frequency at glutamatergic or GABAergic synapse (Chen et al., 2017; Luo et al., 2020). To test whether the increase in sIPSC frequency may be caused by an increase in the number of glycinergic synapses, we performed immunostaining of brainstem sections containing the LSO with antibodies specific for GlyT2 or VGAT, synaptic markers for the glycinergic neurons. Consistent with the increased frequency of sIPSCs, we found a significant increase in the density of both GlyT2 (**Figure 3e-f**) and VGAT (**Figure 3g-h**) staining, suggesting more abundant glycinergic terminals at the LSO of TKO mice. Together, these data show that the pan-neurexin deletion in MNTB neurons has surprisingly increased the baseline glycinergic neurotransmission, which is distinct from the phenotypes observed in glutamatergic and GABAergic neurons (Chen et al., 2017; Luo et al., 2021).

## Deletion of neurexins impairs the tight coupling between voltage-gated $\text{Ca}^{2+}$ channels and synaptic vesicles at the glycinergic synapse

Because neurexins are shown to exert diverse impacts including specifying the function of voltage-gated  $\text{Ca}^{2+}$  channels and their spatial clustering within the presynaptic active zone (Chen et al., 2017; Luo et al., 2020; Missler et al., 2003), we approached the matter by first exploring the effect of pan-neurexin deletion on the calcium sensitivity of neurotransmitter



**Figure 3**

### Deletion of neuroligins increases sIPSC frequency and glycinergic synapse density of LSO neurons.

(a) Representative sIPSC traces of LSO neurons. (b) Summary of sIPSC amplitude and frequency. Ctrl (n = 14), TKO (n = 9), P = 0.057, P = 0.015, unpaired two-sided t-test. (c) Representative sIPSC waveforms of LSO neurons. (d) Summary of sIPSC rise time and decay time constant. Ctrl (n = 14), TKO (n = 9), P = 0.376, P = 0.82, unpaired two-sided t-test. (e) Representative confocal microscopy images of MNTB-LSO-containing brainstem slice with specific labeling of GlyT2 (red) and EGFP (green) from both control and Nrnx123 TKO mice at P14. Scale bar, 10  $\mu$ m. (f) Summary of glycinergic synaptic density quantified by GlyT2 immunostaining. Ctrl (n = 42), TKO (n = 42), P = 0.0001, unpaired two-sided t-test. (g-h), Same as (e-f) except for specific labeling of VGAT (red) and EGFP (green). Ctrl (n = 42), TKO (n = 42) P = 0.0047, unpaired two-sided t-test. Data are means  $\pm$  SEM. Number of neuron or sections/animals for immunostaining are indicated in the bars (b, d, f and h). Statistical differences were assessed by Student's t test. (\*P < 0.05, \*\*P < 0.01, \*\*\*P < 0.001). Source data are provided as a Source Data file.

release at the MNTB-LSO synapse. We perfused the brain slice with low  $\text{Ca}^{2+}$  (0.2 mM) ACSF while recording IPSCs continuously. In both control and *Nrxn123* TKO mice, the perfusion of low  $\text{Ca}^{2+}$  ACSF gradually decreased the amplitude of IPSCs (**Figure 4a, c**). However, the reduction rate was much faster in TKO mice such that IPSC amplitudes dropped more than 50% in TKO vs ~10% in control when extracellular  $[\text{Ca}^{2+}]$  reached to approximately 1.5 mM (**Figure 4b, d**). The extracellular  $[\text{Ca}^{2+}]$  blocking 50% of IPSCs were estimated as 1.1 mM and 1.5 mM respectively for control and TKO mice, which are significantly different (**Figure 4e, f**), suggesting that neurexins indeed are crucial for regulating calcium sensitivity of transmitter release at the glycinergic synapse.

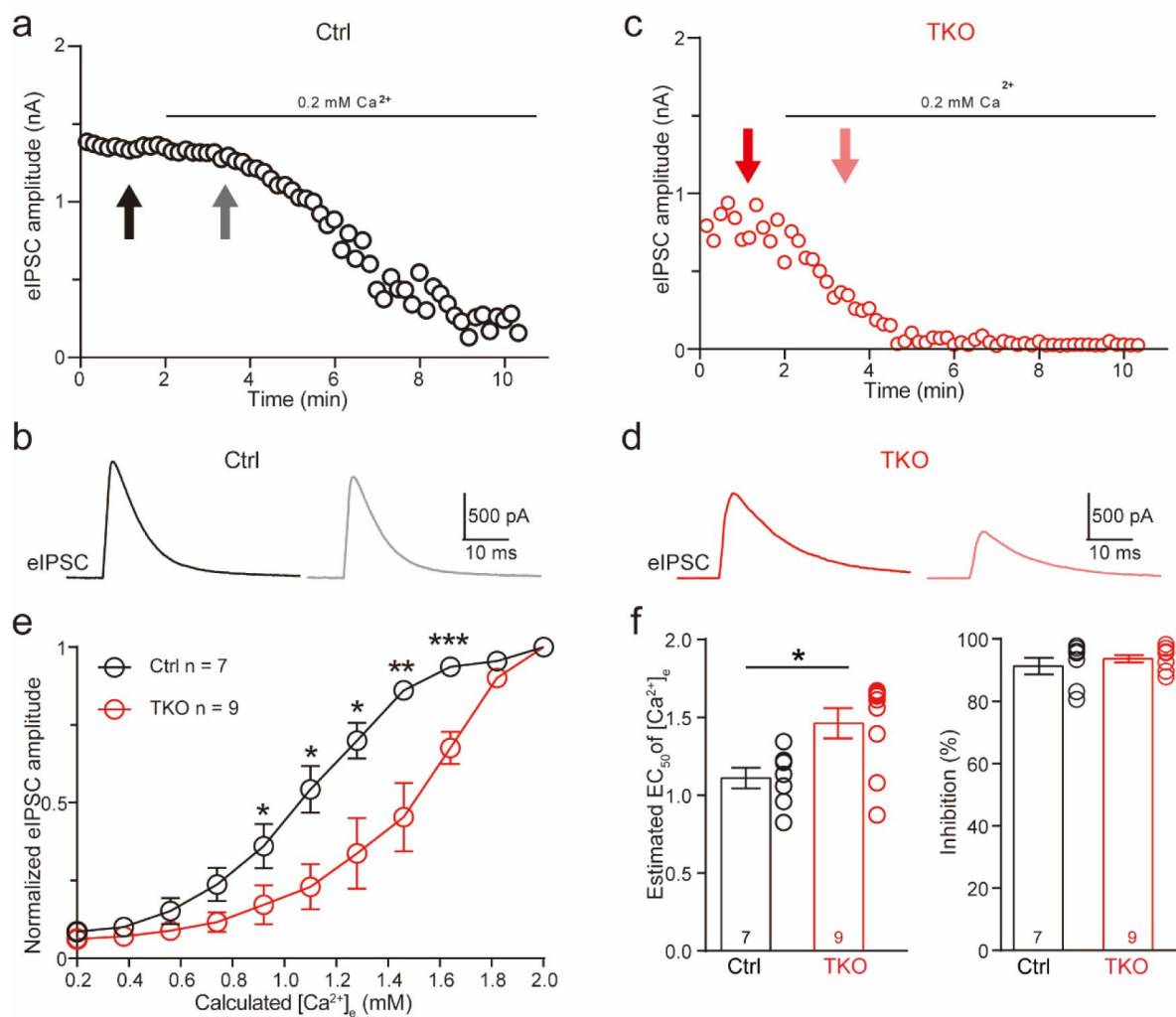
One of the critical factors affecting the calcium sensitivity of release is the spatial distance between  $\text{Ca}^{2+}$  channels and synaptic vesicles (Eggermann et al., 2011; Luo et al., 2020). Additionally, neurexins have been shown to promote tight coupling of  $\text{Ca}^{2+}$  channels with synaptic vesicle at the glutamatergic calyx of Held synapse (Luo et al., 2020). Because the MNTB-LSO glycinergic synapse is similar on multiple aspects of the functional characteristics, being fast, robust, and precise, to the calyx of Held synapse, we tested whether neurexins are also required for the tight coupling of  $\text{Ca}^{2+}$  channels with synaptic vesicle. We incubated the brain slice with high-concentration EGTA-AM and continuously recorded IPSCs from individual LSO neurons. As illustrated in **Figure 5**, EGTA-AM treatment slightly blocked IPSCs at control synapses, suggesting the relative ineffectiveness of EGTA in interfering neurotransmitter release (**Figure 5a, b**). By contrast, at the neurexin-deficient synapses, EGTA-AM treatment remarkably impaired IPSCs (**Figure 5c, d**). On average, the blocking effects of EGTA at neurexin-deficient synapses ( $53.2 \pm 5.4\%$ ) were much stronger than the effect at control synapses ( $15.6 \pm 4.3\%$ ). Therefore, consistent with previous studies at the calyx of Held synapse (Luo et al., 2020), our data provide further support that neurexins are key synaptic organizers for tightly clustering voltage-gated  $\text{Ca}^{2+}$  channels with synaptic vesicles at the glycinergic synapse.

## Discussion

Using the MNTB-LSO synapse as a model, here we provide first assessment of the role of neurexins at glycinergic synapses that mediate prominent synaptic inhibition in the brainstem. Our findings in general strongly corroborate the idea that neurexins are central presynaptic organizers in all major types of fast chemical synapses including glutamatergic, GABAergic and glycinergic.

Specifically, we have shown that various neurexin genes express concurrently in these glycinergic neurons but the expression level of each appears to vary significantly (**Figure 1**). This finding is consistent with many studies in glutamatergic and GABAergic neurons, which converge to show that multiple genes, isoforms, and splicing sites of neurexins normally co-express in individual neurons and the expression profiles are highly cell-type specific and brain region specific (Fuccillo et al., 2015; Schreiner et al., 2014; Treutlein et al., 2014; Uchigashima et al., 2019). Distinct neurexins may interact separately and/or redundantly with diverse signaling molecules to form specific molecular codes for shaping neuronal connectivity and functional properties (Aoto et al., 2015; Aoto et al., 2013; Dai et al., 2019; Lin et al., 2023; Lloyd et al., 2023; Sudhof, 2017). The observed differential expression of individual neurexins suggests the possibility that specific isoforms may exert dominant regulatory roles in neurotransmission in different synapses. For instance, compelling evidence indicates that *Nrxn3*, arguably the best studied neurexin isoform, is selectively required for intact synaptic function in the olfactory bulb GABAergic neurons (Aoto et al., 2015; Trotter et al., 2023). Understanding the distinct roles of each neurexin isoform in regulating glycinergic synaptic transmission is crucial for future studies. Particularly noteworthy is the potential involvement of *Nrxn3*, the most abundant isoform, in this synapse.



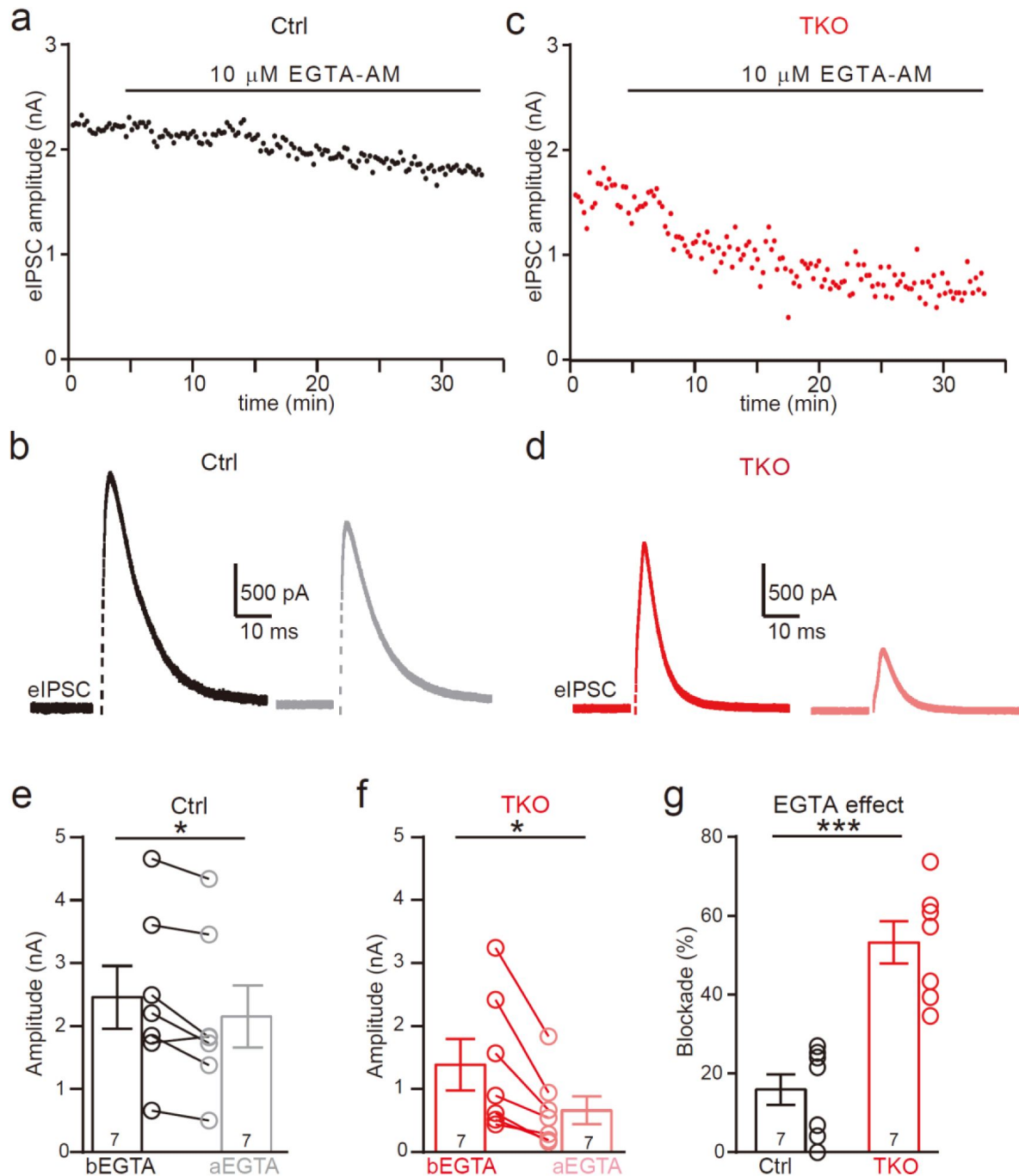


**Figure 4**

### Deletion of neurexins increases the $\text{Ca}^{2+}$ sensitivity of release at the glycinergic synapse.

(a) Representative time courses of IPSC amplitudes for individual LSO neurons of control mice during the perfusion of low  $\text{Ca}^{2+}$  ACSF. (b) Representative IPSC traces of control mice before and during low  $\text{Ca}^{2+}$  perfusion at the indicated points in a. (c & d) Same as a & b, except of TKO mice. (e) Summary of normalized IPSC amplitudes plotted against the calculated extracellular  $\text{Ca}^{2+}$  concentration for control and TKO mice. Ctrl (n = 7), TKO (n = 9),  $P = 0.0311$  for  $0.92 \text{ mM Ca}^{2+}$ ,  $P = 0.0104$  for  $1.10 \text{ mM Ca}^{2+}$ ,  $P = 0.0208$  for  $1.28 \text{ mM Ca}^{2+}$ ,  $P = 0.0067$  for  $1.46 \text{ mM Ca}^{2+}$ ,  $P = 0.0007$  for  $1.64 \text{ mM Ca}^{2+}$ , unpaired two-sided t-test. (f) Summary of  $\text{EC}_{50}$  of  $\text{Ca}^{2+}$  for half-blocking of IPSCs and the blocking effect of  $0.2 \text{ mM Ca}^{2+}$  on IPSC amplitude. Ctrl (n = 7), TKO (n = 9),  $P = 0.0139$ ,  $P = 0.3985$ , unpaired two-sided t-test.

Data are means  $\pm$  SEM. Number of neuron are indicated in the graph (e) or bars (f). Statistical differences were assessed by Student's t test. (\* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ ). Source data are provided as a Source Data file.



**Figure 5**

**Deletion of neurexins increases the blocking effect of EGTA on glycinergic IPSCs.**

(a) Representative time courses of IPSC amplitudes for individual LSO neurons of control mice during the treatment of high concentration of EGTA-AM. (b) Representative IPSC traces of control mice before and after EGTA. (c, d) Same as a, b, except of TKO mice. (e) Summary of IPSC amplitude before and after EGTA treatment of control mice. Ctrl (n = 7), TKO (n = 7),  $P = 0.0201$ , paired two-sided t-test. (f) Summary of IPSC amplitude before and after EGTA treatment of TKO mice. Ctrl (n = 7), TKO (n = 7),  $P = 0.0123$ , paired two-sided t-test. (g) Summary of the blocking percentage of IPSCs by EGTA. Ctrl (n = 7), TKO (n = 7),  $P = 0.0001$ , unpaired two-sided t-test.

Data are means  $\pm$  SEM. Number of neuron are indicated in the bars (e to g). Statistical differences were assessed by Student's t test. (\* $P < 0.05$ , \*\*\* $P < 0.001$ ). Source data are provided as a Source Data file.

We further show that neurexins are critical in controlling the strength and precise timing of the glycinergic synaptic inhibition at LSO neurons by clustering  $\text{Ca}^{2+}$  channels tightly with synaptic vesicles in presynaptic release sites (**Figures 2**, **4** and **5**). These data strongly corroborate two perspectives on the function of neurexins as central presynaptic organizers. First, deletion of all neurexins not only weakens the glycinergic synaptic connectivity but also impairs the temporal precision of synaptic inhibition, exactly as it does at the glutamatergic calyx of Held synapse (Luo *et al.*, 2020). Since both the glutamatergic calyx of Held synapse and the glycinergic MNTB-LSO synapse in the auditory brainstem are extremely strong and fast (Beiderbeck *et al.*, 2018; Grothe, 2003), the orchestrating role of neurexins in controlling the timing and magnitude of their synaptic transmission is therefore desirable for the auditory information processing. Secondly, neurexins execute their role at the glycinergic synapse mainly by tightly organizing  $\text{Ca}^{2+}$  channels within the release sites, again exactly as it does at the calyx of Held synapse (Luo *et al.*, 2020). Nanodomain coupling between  $\text{Ca}^{2+}$  channels and synaptic vesicles has been found in many fast central synapses and is functionally crucial for increasing the efficacy and speed of neurotransmission (Eggermann *et al.*, 2011). This similar role of neurexins in clustering  $\text{Ca}^{2+}$  channels tightly at presynaptic active zones of not only the glutamatergic synapse but also the glycinergic synapse prompts a shared mechanism and further research on the potential molecular candidates.

On the other hand, our data surprisingly reveal that deletion of neurexins may promote the formation/maintenance of the glycinergic synapse, as evidenced by the increase in the frequency of spontaneous synaptic events measured at LSO neurons as well as the increase in the glycinergic synaptic density in *Nrxn123* TKO mice (**Figure 3**). Early studies of TKOs of  $\alpha$ -neurexins or  $\beta$ -neurexins lead to general conclusion that neurexins are not essential for the formation/maintenance of excitatory and inhibitory synapses (Anderson *et al.*, 2015; Missler *et al.*, 2003). Moreover, deletion of all  $\alpha$ -neurexins and  $\beta$ -neurexins has no impact on the formation/maintenance of glutamatergic synapse (Luo *et al.*, 2020) and GABAergic synapse made by somatostatin-positive interneurons (Chen *et al.*, 2017), but produces partial loss (~30%) of GABAergic synapse made by parvalbumin-positive interneurons (Chen *et al.*, 2017). There are several possibilities that may explain our present findings on the glycinergic synapse. First, the most parsimonious explanation would be that neurexins play contradictory dual roles at the glycinergic synapse: orchestrate the synchronized synaptic transmission by promoting tight coupling of  $\text{Ca}^{2+}$  channels with synaptic vesicles at the presynaptic active zone; restrict the formation of weaker synapse or spontaneous transmission. Another well-established example of synaptic protein with dual functions in synaptic transmission is synaptotagmin 1, which acts as a calcium sensor to trigger synchronous release while clamps and restricts spontaneous fusion of vesicles on the other hand (Sudhof, 2013). It is worthy of note that synaptotagmin can specifically interact with neurexins (Hata *et al.*, 1993). Second, deletion of neurexins impairs the strength of the MNTB-LSO synapse, which may indirectly interfere with synaptic elimination process and therefore lead to an increase in synaptic density. Third, the more abundant glycinergic synapses may arise from the developmental compensation of synapse formation by other sources of glycinergic neurons (Jalabi *et al.*, 2013). Strong reduction in the strength of the MNTB-LSO synapse caused by selective deletion of neurexins may shift the winner of synaptic competition toward other parallel but much weaker glycinergic inputs in normal animals.

## Methods

### Mouse breeding and virus injection

All animal experiments were approved by the Institutional Animal Care and Use Committee at Bioland Laboratory, Guangzhou National Laboratory, Guangzhou, China. Transgenic *Nrxn123* cTKO mice of either sex were used for all experiments. Mice were housed at room temperature on a 12 h light–dark cycle (7:00 to 19:00, light) with food and water freely available.

To delete neurexins in MNTB neurons, we injected P0 mice with AAV2/9-hSyn-EGFP-P2A-Cre-WPRE-hGH-Poly (Cat:PT-0156, BrainVTA Co., China). Mice were anesthetized on ice and immediately transferred to a stereotaxic apparatus (RWD Life Technology, China). Injection was achieved by connecting a glass pipette with tip diameter 20-30  $\mu\text{m}$  to Nanoject III programmable nanoliter injector (Drummond Scientific Company, USA). To enhance virus infection efficiency, we decreased the dosage per injection while increasing the frequency of injections. Additionally, we ensured the pipette remained immobilized for 20-30 seconds to guarantee virus absorption at injection sites. As a result of this strategy, we estimated that the vast majority of MNTB neurons were inoculated by AAVs. Bilateral injections were performed at the following coordinates: AP from the most rostral point: 5.61 mm, ML:  $\pm 0.27$  mm, DV: 3.58 mm. For optogenetics experiments, we injected 200 nl (AAV2/9-hSyn-EGFP-P2A-Cre-WPRE-hGH-PolyA and AAV2/9-EF1 $\alpha$ -DIO-hChR2(H134R)-mCherry-WPRE-hGH-Poly, 1:1, Cat: PT-0002, BrainVTA Co., China) or 100 nl of AAV2/9-hSyn-hChR2(H123R)-EYFP-WPRE-hGH-PolyA (Cat: PT-1317, BrainVTA Co., China) into each-side MNTB. P13-14 mice of either sex were used for most experiments including electrophysiology, FISH and immunohistochemistry.

## Acute slice preparation for electrophysiology

Coronal brain slices containing both the MNTB and LSO were prepared similarly as described previously. In brief, mice of P13-14 were decapitated; brains were rapidly isolated and fixed on the cutting platform a vibratome (VT1200s; Leica, USA), which was immersed in oxygenated cold ACSF containing (in mM): 119 NaCl, 26 NaHCO<sub>3</sub>, 10 glucose, 1.25 NaH<sub>2</sub>PO<sub>4</sub>, 2.5 KCl, 0.05 CaCl<sub>2</sub>, 3 MgCl<sub>2</sub>, 2 Na-pyruvate, and 0.5 ascorbic acid, pH 7.4. Transverse slices of 250  $\mu\text{m}$  were sectioned and transferred into a beaker with bubbled ACSF containing (in mM): 119 NaCl, 26 NaHCO<sub>3</sub>, 10 glucose, 1.25 NaH<sub>2</sub>PO<sub>4</sub>, 2.5 KCl, 2 CaCl<sub>2</sub>, 1 MgCl<sub>2</sub>, 2 Na-pyruvate, and 0.5 ascorbic acid, pH 7.4. Slices were recovered at 35°C for 30 minutes, and stored at room temperature ( $\sim 21$ -23°C) for experiments.

## Whole-cell voltage-clamped recordings

Whole-cell voltage-clamped recordings were made from visually identified LSO neurons in acute brain slices using an Axopatch 700B amplifier (Molecular Devices, USA) and pClamp 11 software (Molecular Devices, USA). Patch pipettes were pulled from borosilicate glass capillaries using a two-stage vertical puller (PC-100, Narashige, Japan). Pipette resistance was 4-6 M $\Omega$  when filled with an internal solution containing (in mM): 145 K-gluconate, 6 KCl, 10 HEPES, 3 Na<sub>2</sub>-phosphocreatine, 4 Mg-ATP, 0.3 Na<sub>2</sub>-GTP, 0.2 EGTA, 2 Qx-314; pH 7.2 with KOH. With the extracellular [Cl<sup>-</sup>] of 133 mM and a temperature of 22°, the equilibrium potential of the Cl<sup>-</sup> was approximately -78.8 mV. In voltage-clamp configuration, IPSCs were recorded by holding cells at 10 mV in ACSF containing CNQX (20  $\mu\text{M}$ ), D-AP5 (50  $\mu\text{M}$ ), to block AMPA, NMDA receptors, respectively. For some experiments, 2  $\mu\text{M}$  strychnine, a potent glycine receptor antagonist, was added to ACSF to verify that IPSCs recorded at LSO neurons are glycinergic. Membrane potentials were not corrected for liquid junction potential. Electrical fiber stimulation was performed using a concentric bipolar electrode (CBAEB75, FHC Inc, USA) positioned medially to the LSO and MNTB. Short pulses (0.1 ms) were delivered via an isolated stimulator (Model 2100; A-M Systems, USA). For optogenetics, a light stimulation of 5 ms was delivered by LED (pE-300 white, CoolLED, UK).

## Whole-cell current-clamp recordings

Whole-cell current-clamped recordings were performed similarly as voltage-clamped recordings, except that the internal solution did not contain Qx-314. Action potentials at MNTB neurons were recorded by holding the membrane potential at approximately -70 mV. For optogenetics, a light stimulation of 5 ms was delivered by LED (pE-300 white, CoolLED, UK).

## RNAscope fluorescent in-situ hybridization (FISH) and immunohistochemistry

Mice were anesthetized and perfused at speed of 1ml/min with 1×PBS for 10 min followed by 4% paraformaldehyde (PFA) for 10 min. Brains were removed and stored in 4% PFA overnight at 4°C. After going through a 10%, 20%, and 30% sucrose gradient for cryo-protection, the tissue was embedded in OCT (Cat: 14020108926, Leica) and rapidly frozen on dry ice. Transverse sections at 16 µm were cut at -20 °C using a cryostat (CM3050-S, Leica) and mounted directly on Superfrost Plus slides (Cat: PRO-04, Matsunami Platinum Pro). FISH and immunohistochemistry were performed using the RNAscope multiplex platform (Multiplex Reagent Kit, Cat: 323100; Co-detection ancillary Kit, Cat: 323180, Advanced Cell Diagnostics) following the manufacturer's instructions. RNA probes for *Nrxn1*, *Nrxn2*, and *Nrxn3* (Cat: 461511-C3, 533531-C2, 505431 respectively, Advanced Cell Diagnostics) and primary antibody against VGLUT1 (guinea pig, polyclonal, 1:500, Millipore) were used. Secondary antibodies were Alexa Fluor conjugates (1:400; Thermo Fisher). Samples were mounted with a Vectashield hard-set antifade mounting medium (Cat: H-1500, Vector Laboratories). Images were acquired using Zeiss LSM800 confocal microscope with a 63x oil-immersion objective (1.4 numerical aperture) or OLYMPUS IXplore SpinSR confocal microscope with a 60x oil-immersion objective (1.5 numerical aperture). Image analysis were performed using Zeiss 2.6, cellSens Dimension 3.2 and Image J.

## Pharmacological manipulations

To estimate the calcium sensitivity of neurotransmitter release, we perfused the slice with ACSF containing low  $\text{Ca}^{2+}$  (0.2 mM) and continuously recorded eIPSC.  $\text{EC}_{50}$  of  $\text{Ca}^{2+}$  sensitivity was determined using a sigmoidal function ( $Y = \text{Min} + (\text{Max} - \text{Min}) / (1 + 10^{((\text{LogEC}_{50} - X) * n)))$ ), where Y represents IPSC amplitude, X denotes the calculated extracellular  $\text{Ca}^{2+}$  concentration, Min refers to the minimal amplitude, Max represents the maximum amplitude. To estimate the spatial relationship between voltage-gated  $\text{Ca}^{2+}$  channels and synaptic vesicles, we perfused the slice with ACSF containing 20 µM EGTA-AM for 25~30 mins while continuously recording eIPSC. The blocking effect of EGTA on eIPSCs were evaluated by comparing eIPSCs before and after EGTA treatment.

## Statistical analysis

Data were analyzed in the IgorPro (Version 8) and GraphPad Prism 9 software. Summary of data was displayed as mean ± SEM. Statistical analysis was performed using Student's *t* test, two-way ANOVA, and Mann-Whitney U test. Significance of difference was accepted at  $P < 0.05$ .

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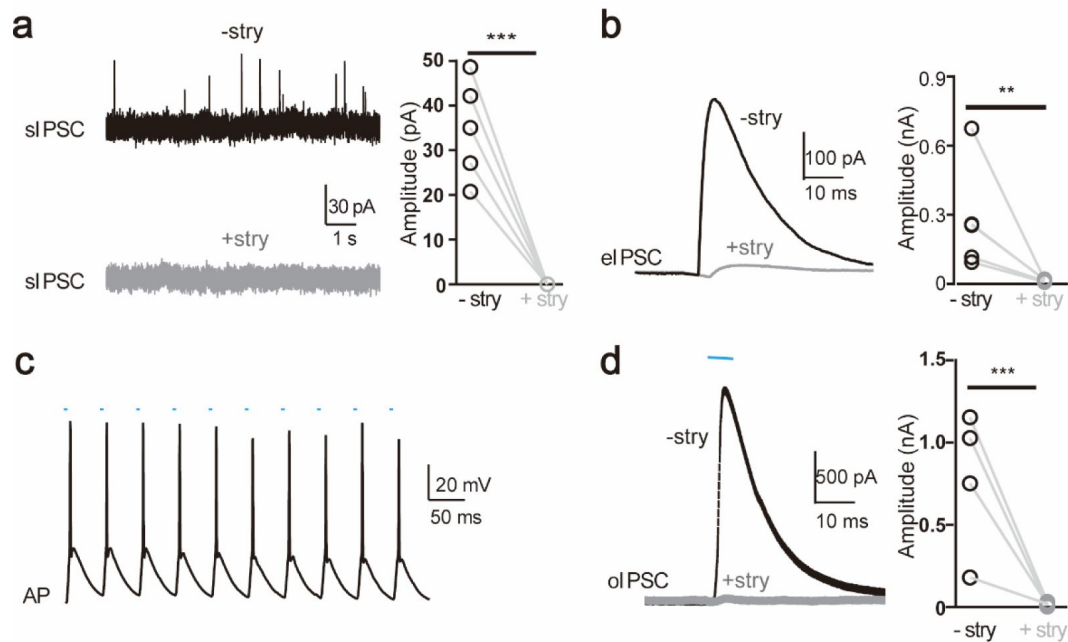
## Author contributions

Conceptualization, F.L.; Experiments, H.J., R.X., X.N., Z.S., X.X., R. P.; Data analysis, H.J., R.X., X.N., Z.S., X.X., Y.Z., F.L.; Writing – Original Draft, F.L.; Writing – Review & Editing, H.J., Y.Z., F.L.



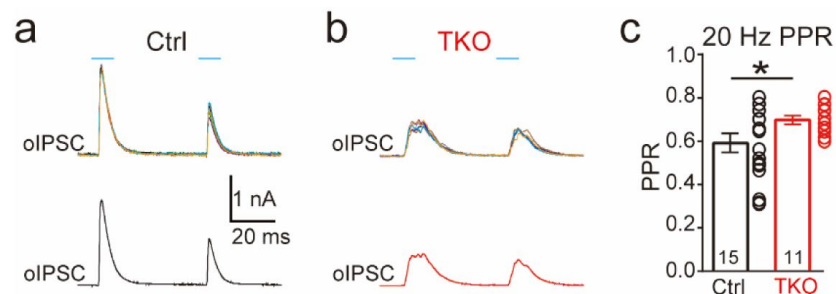
## Declaration of interest

The authors declare no competing financial interests.



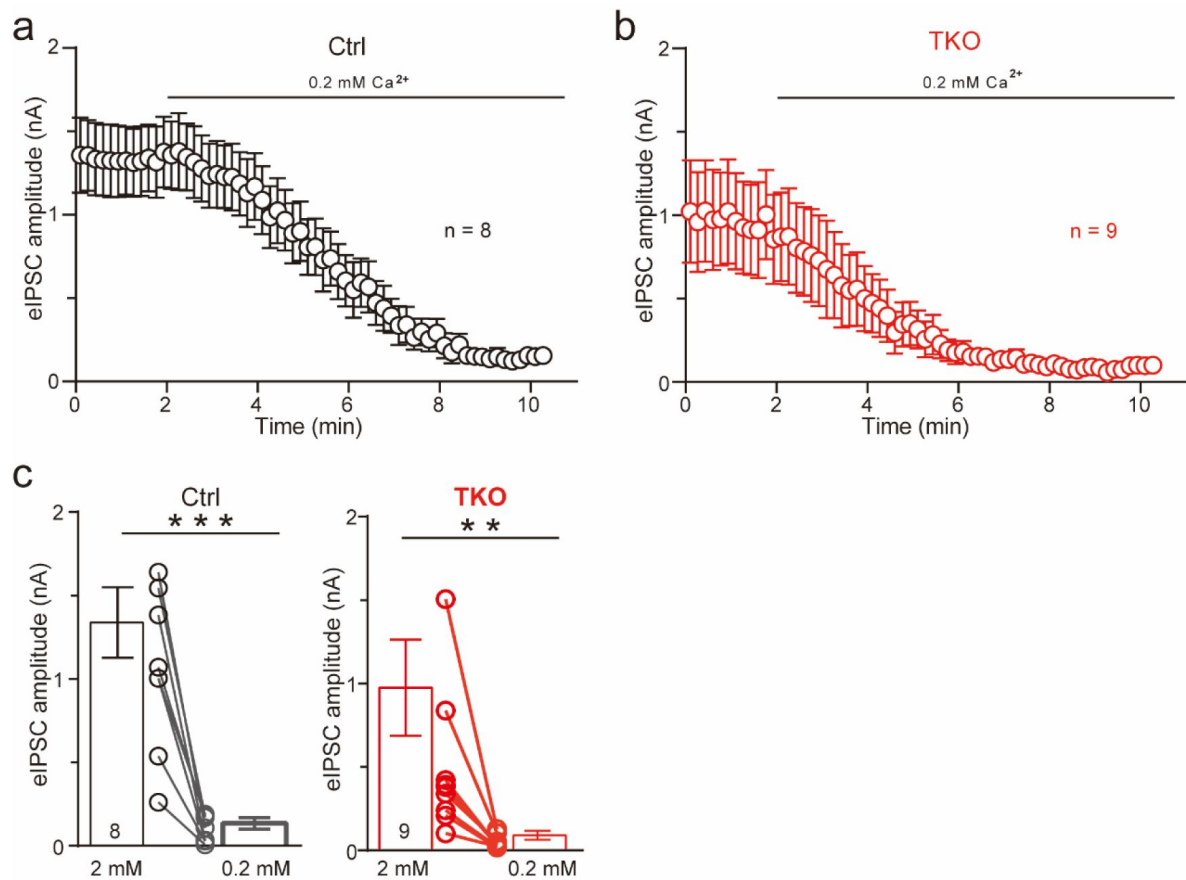
**Figure S1**

**a** Representative sIPSC traces of LSO neurons before and after application of potent glycine receptor blocker strychnine (stry) (left) and summary of sIPSC amplitude without and with stry (right).  
**b** Representative eIPSC traces of LSO neurons before and after stry application (left) and summary of eIPSC amplitude without and with stry (right).  
**c** Representative action potential (AP) firing of MNTB neurons infected with Chr2 during high frequency repetitive pulse stimulation of 473 nm.  
**d** Representative oIPSC traces of LSO neurons before and after stry application (left) and summary of oIPSC amplitude without and with stry (right).



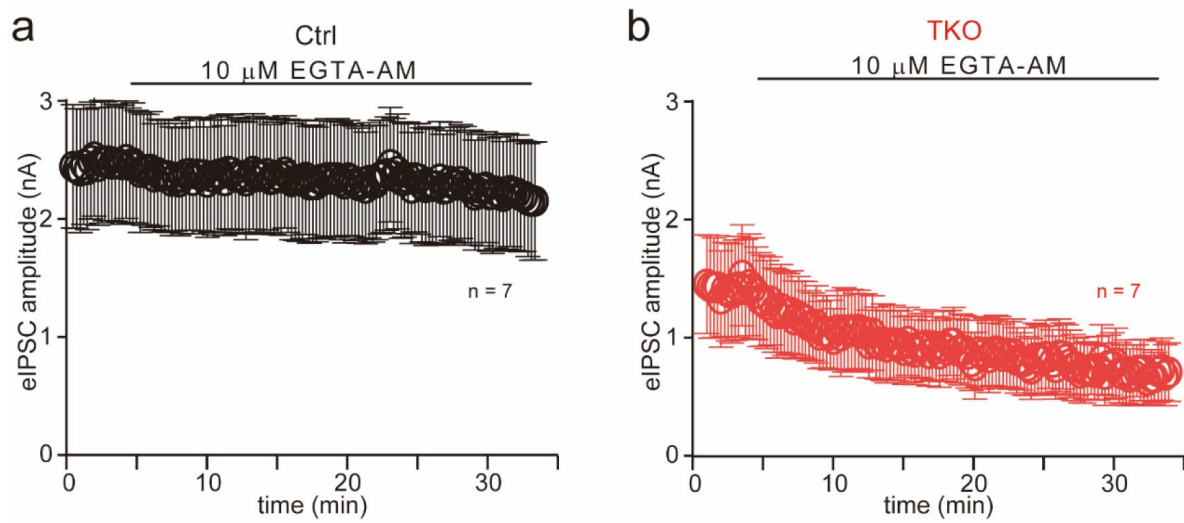
**Figure S2**

**a,b** Representative IPSC traces of LSO neurons evoked by paired-pulse (20 Hz) optogenetic stimulation of Chr2 expressed in MNTB neurons in control (**a**) and Nr1n TKO (**b**) mice. Top, individual trials. Bottom, averaged trace of IPSC.  
**c** Summary of paired-pulse ratio (PPR) of IPSC amplitude.



**Figure S3**

**a** Summary of eIPSC amplitude of control during 0.2 mM  $\text{Ca}^{2+}$  perfusion. Mean  $\pm$  SEM.  
**b** Summary of eIPSC amplitude of TKO during 0.2 mM  $\text{Ca}^{2+}$  perfusion. Mean  $\pm$  SEM.  
**c** Average of eIPSC amplitude before and after 0.2 mM  $\text{Ca}^{2+}$  perfusion. Mean  $\pm$  SEM.  
 (\*\*P < 0.01, \*\*\*P < 0.001), paired t-test.



**Figure S4**

**a** Summary of eIPSC amplitude of control during 10  $\mu$ M EGTA-AM treatment. Mean  $\pm$  SEM.

**b** Summary of eIPSC amplitude of TKO during 10  $\mu$ M EGTA-AM treatment. Mean  $\pm$  SEM

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### Reviewer #1 (Public Review):

Jiang et al. demonstrated that ablating Neurexins results in alterations to glycinergic transmission and its calcium sensitivity, utilizing a robust experimental system. Specifically, the authors employed rAAV-Cre-EGFP injection around the MNTB in *Nrxn1/2/3* triple conditional mice at P0, measuring Glycine receptor-dependent IPSCs from postsynaptic LSO neurons at P13-14. Notably, the authors presented a clear reduction of 60% and 30% in the amplitudes of opto- and electric stimulation-evoked IPSCs, respectively. Additionally, they observed changes in kinetics, alterations in PPR, and sensitivity to lower calcium and the calcium chelator, EGTA, indicating solid evidence for changes in presynaptic properties of glycinergic transmission.

Furthermore, the authors uncovered an unexpected increase in sIPSC frequency without altering amplitude. Although the precise mechanism remains unknown, the authors discussed this complex phenotype by considering various possibilities, including the potential scenario where the augmentation in synapses may result from *Nrxn* deletion rather than being a causal effect.

<https://doi.org/10.7554/eLife.94315.2.sa2>

### Reviewer #2 (Public Review):

Summary:

In this manuscript, Jiang et al., explore the role of neurexins at glycinergic MNTB-LSO synapses. The authors utilize elegant and compelling ex vivo slice electrophysiology to assess how the genetic conditional deletion of *Nrxns1-3* impacts inhibitory glycinergic synaptic transmission and found that TKO of neurexins reduced electrically and optically evoked IPSC amplitudes, slowed optically evoked IPSC kinetics and reduced presynaptic release probability. The authors use classic approaches including reduced  $[Ca^{2+}]$  in ACSF and EGTA chelation to propose that changes in these evoked properties are likely driven by the loss of calcium channel coupling. Intriguingly, while evoked transmission was impaired, the authors reported that spontaneous IPSC frequency was increased, due to an increase in the number of synapses in LSO. Overall, this manuscript provides important insight into the role of neurexins at the glycinergic MNTB-LSO synapse and further emphasizes the need for continued study of both the non-redundant and redundant roles of neurexins.

The authors have addressed all of my previous concerns.

<https://doi.org/10.7554/eLife.94315.2.sa1>

**Reviewer #3 (Public Review):****Summary:**

The authors investigate the hypothesis that neurexins serve a crucial role as regulators of the synaptic strength and timing at the glycinergic synapse between neurons of the medial nucleus of the trapezoid body (MNTB) and the lateral superior olivary complex (LSO). It is worth mentioning that LSO neurons are an integration station of the auditory brainstem circuit displaying high reliability and temporal precision. These features are necessary for computing interaural cues to derive sound source location from comparing the intensities of sounds arriving at the two ears. In this context, the authors' findings build up according to the hypothesis first by displaying that neurexins were expressed in the MNTB at varying levels. They followed this up with deletion of all neurexins in the MNTB through the employment of a triple knock-out (TKO). Using electrophysiological recordings in acute brainstem slices of these TKO mice, they gathered solid evidence for the role of neurexins in synaptic transmission at this glycinergic synapse primarily by ensuring tight coupling of Ca<sup>2+</sup> channels and vesicular release sites. Additionally, the authors uncovered a connection between the deletion of neurexins and a higher number of glycinergic synapses of TKO mice, for which they provided evidence in the form of immunostainings and related it to electrophysiological data on spontaneous release. Consequently, this investigation expands our knowledge on the molecular regulation of synaptic transmission at glycinergic synapses, as well as on the auditory processing at the level of the brainstem.

**Strengths:**

The authors demonstrate substantial results in support of the hypothesis of a critical role of neurexins for regulating glycinergic transmission in the LSO using various techniques. They provide evidence for the expression of neurexins in the MNTB and consecutively successfully generate and characterize the neurexin TKO. For their study on LSO IPSCs the authors transduced MNTB neurons by co-injection of virus carrying Cre and ChR2 and subsequently optogenetically evoke release of glycine. As a result, they observed a significant reduction in amplitude and significantly slower rise and decay times of the IPSCs of the TKO in comparison with control mice in which MNTB neurons were only transduced with ChR2. Furthermore, they observed an increased paired pulse ratio (PPR) of LSO IPSCs in the TKO mice, indicating lower release probability. Elaborating on the hypothesis that neurexins are essential for the coupling of synaptic vesicles to Ca<sup>2+</sup> channels, the authors show lowered Ca<sup>2+</sup> sensitivity in the TKO mice. Additionally, they reveal convincing evidence for the connection between the increased frequency of spontaneous IPSC and the higher number of glycinergic synapses of the LSO in the TKO mice, revealed by immunolabeling against the glycinergic presynaptic markers GlyT2 or VGAT.

**Weaknesses:**

A concern is on novelty as this work on the effects of pan-neurexin deletion in a glycinergic synapse is quite consistent with the authors prior work on glutamatergic synapses (Luo et al., 2020).

<https://doi.org/10.7554/eLife.94315.2.sa0>

**Author response:**

The following is the authors' response to the original reviews.

### **eLife assessment**

*This study provides important insights into the role of neurexins as regulators of synaptic strength and timing at the glycinergic synapse between neurons of the medial nucleus of the trapezoid body and the lateral superior olive, key components of the auditory brainstem circuit involved in computing sound source location from differences in the intensity of sounds arriving at the two ears. Through an elegant combination of genetic manipulation, fluorescence in-situ hybridization, ex vivo slice electrophysiology, pharmacology, and optogenetics, the authors provide convincing evidence to support their claims. While further work is needed to reveal the mechanistic basis by which neurexins influence glycinergic neurotransmission, this work will be of interest to both auditory and synaptic neuroscientists.*

We appreciate the recognition of the significance of our study in shedding light on the role of neurexins in regulating synaptic strength and timing at the glycinergic synapse. Indeed, further investigations are warranted to delve deeper into the specific role of each different variant of neurexins in the future. We hope that our work will spark more interest and collaboration in unraveling the complexities of molecular codes of synaptic function.

### **Public Reviews:**

#### **Reviewer #1 (Public Review):**

*Jiang et al. demonstrated that ablating Neurexins results in alterations to glycinergic transmission and its calcium sensitivity, utilizing a robust experimental system. Specifically, the authors employed rAAV-Cre-EGFP injection around the MNTB in *Nrxn1/2/3* triple conditional mice at P0, measuring Glycine receptor-dependent IPSCs from postsynaptic LSO neurons at P13-14. Notably, the authors presented a clear reduction of 60% and 30% in the amplitudes of opto- and electric stimulation-evoked IPSCs, respectively. Additionally, they observed changes in kinetics, alterations in PPR, and sensitivity to lower calcium and the calcium chelator, EGTA, indicating solid evidence for changes in presynaptic properties of glycinergic transmission.*

*Furthermore, the authors uncovered an unexpected increase in sIPSC frequency without altering amplitude. Despite the reduction in evoked IPSC, immunostaining revealed an increase in GlyT2 and VGAT in TKO mice, supporting the notion of an increase in synapse number. However, the reviewer expresses caution regarding the authors' conclusion that "glycinergic neurotransmission likely by promoting the synapse formation/maintenance, which is distinct from the phenotypes observed in glutamatergic and GABAergic neurons (Chen et al., 2017; Luo et al., 2021)", as outlined in lines 173-175. The reviewer suggests that this statement may be overstated, pointing out the authors' own discussion in lines 254-265, which acknowledges multiple possibilities, including the potential that the increase in synapses is a consequence rather than a causal effect of *Nrxn* deletion.*

We appreciate the reviewer's thoughtful evaluation of our study. We agree that our conclusion regarding the promotion of synapse formation/maintenance may have been overstated and recognize the need for a more nuanced interpretation of our findings. Accordingly, we have revised our interpretation by discussing carefully the various possibilities that may cause the observed increase in synapse number in line 256-266.

#### **Reviewer #2 (Public Review):**

*Summary:*



*In this manuscript, Jiang et al., explore the role of neurexins at glycinergic MNTB-LSO synapses. The authors utilize elegant and compelling ex vivo slice electrophysiology to assess how the genetic conditional deletion of *Nrxns1-3* impacts inhibitory glycinergic synaptic transmission and found that TKO of neurexins reduced electrically and optically evoked IPSC amplitudes, slowed optically evoked IPSC kinetics and reduced presynaptic release probability. The authors use classic approaches including reduced  $[Ca^{2+}]$  in ACSF and EGTA chelation to propose that changes in these evoked properties are likely driven by the loss of calcium channel coupling. Intriguingly, while evoked transmission was impaired, the authors reported that spontaneous IPSC frequency was increased, potentially due to an increased number of synapses in LSO. Overall, this manuscript provides important insight into the role of neurexins at the glycinergic MNTB-LSO synapse and further emphasizes the need for continued study of both the non-redundant and redundant roles of neurexins.*

We thank the reviewer for the strong comments and support of our work.

*Strengths:*

*This well-written manuscript seamlessly incorporates mouse genetics and elegant ex vivo electrophysiology to identify a role for neurexins in glycinergic transmission at MNTB-LSO synapses. Triple KO of all neurexins reduced the amplitude and timing of evoked glycinergic synaptic transmission. Further, spontaneous IPSC frequency was increased. The evoked synaptic phenotype is likely a result of reduced presynaptic calcium coupling while the spontaneous synaptic phenotype is likely due to increased synapse numbers. While neuroligin-4 has been identified at glycinergic synapses, this study, to the best of my knowledge, is the first to study *Nrxn* function at these synapses.*

We again appreciate the positive feedback on the strengths of our study. We agree that the observed reduction in evoked synaptic transmission and the increase in spontaneous IPSC frequency provide intriguing insights into the function of neurexins in regulating glycinergic synaptic activity.

*Weaknesses:*

*The data are compelling and report an intriguing functional phenotype. The role of Neurexins redundantly controls calcium channel coupling has been previously reported. Mechanistic insight would significantly strengthen this study.*

We wholeheartedly agree with the reviewer that understanding how neurexins control calcium channel coupling at the presynaptic active zone is crucial for elucidating their role in synaptic transmission. While our current study has provided compelling evidence for the functional phenotypes of pan-neurexin deletion, we recognize the importance of investigating the underlying molecular mechanisms in future research. Exploring these mechanisms would undoubtedly enhance our understanding of neurexin function at various synapses and contribute to advancing the field.

*The claim that triple KO of *Nrxns* from MNTB increases the number of synapses in LSO is not strongly supported.*

We agree. Echoing the suggestion made by reviewer 1 (as mentioned above), we acknowledge that the claim regarding the increase in synapse numbers in the LSO following the triple knockout of neurexins from the MNTB was overstated. Consequently, we have revised our conclusions more carefully to reflect this adjustment.

*Despite the stated caveats of measuring electrically evoked currents and the more robust synaptic phenotypes observed using optically evoked transmission, the authors rely heavily on electrical stimulation for most measurements.*

We acknowledge that optogenetic stimulation offers crucial advantages, and we have provided a balanced discussion of the caveats associated with both methods in our manuscript. Additionally, we have conducted new optogenetic experiments specifically for measuring the paired-pulse ratio in control and Nrnx123 TKO mice. These results have been included as a new supplementary figure (Figure S2).

For experiments involving EGTA and low  $\text{Ca}^{2+}$  manipulations, we opted for electrical stimulation due to concerns regarding potential side effects of optogenetics, including the phototoxicity and photobleaching during prolonged light exposure.

*The differential expression of individual neurexins might indicate that specific neurexins may dominantly regulate synaptic transmission, however, this possibility is not discussed in detail.*

We thank the reviewer for bringing up this important point. The differential expression of individual neurexins indeed suggests that specific neurexins may play dominant roles in regulating synaptic transmission. While our study primarily focused on the collective impact of ablating all neurexins, we acknowledge the significance of exploring the specific contributions of individual neurexin isoforms in the future. Understanding the distinct roles of each neurexin isoform could provide valuable insights into the precise mechanisms underlying synaptic function and plasticity. We have added discussion in our revised manuscript Line223-230.

#### **Reviewer #3 (Public Review):**

##### **Summary:**

*The authors investigate the hypothesis that neurexins serve a crucial role as regulators of the synaptic strength and timing at the glycinergic synapse between neurons of the medial nucleus of the trapezoid body (MNTB) and the lateral superior olivary complex (LSO). It is worth mentioning that LSO neurons are an integration station of the auditory brainstem circuit displaying high reliability and temporal precision. These features are necessary for computing interaural cues to derive sound source location from comparing the intensities of sounds arriving at the two ears. In this context, the authors' findings build up according to the hypothesis first by displaying that neurexins were expressed in the MNTB at varying levels. They followed this up with the deletion of all neurexins in the MNTB through the employment of a triple knock-out (TKO). Using electrophysiological recordings in acute brainstem slices of these TKO mice, they gathered solid evidence for the role of neurexins in synaptic transmission at this glycinergic synapse primarily by ensuring tight coupling of  $\text{Ca}^{2+}$  channels and vesicular release sites. Additionally, the authors uncovered a connection between the deletion of neurexins and a higher number of glycinergic synapses in TKO mice, for which they provided evidence in the form of immunostainings and related it to electrophysiological data on spontaneous release. Consequently, this investigation expands our knowledge on the molecular regulation of synaptic transmission at glycinergic synapses, as well as on the auditory processing at the level of the brainstem.*

##### **Strengths:**

*The authors demonstrate substantial results in support of the hypothesis of a critical role of neurexins for regulating glycinergic transmission in the LSO using various techniques.*

*They provide evidence for the expression of neurexins in the MNTB and consecutively successfully generate and characterize the neurexin TKO. For their study on LSO IPSCs the authors transduced MNTB neurons by co-injection of virus-carrying Cre and ChR2 and subsequently optogenetically evoke release of glycine. As a result, they observed a significant reduction in amplitude and significantly slower rise and decay times of the IPSCs of the TKO in comparison with control mice in which MNTB neurons were only transduced with ChR2. Furthermore, they observed an increased paired pulse ratio (PPR) of LSO IPSCs in the TKO mice, indicating lower release probability. Elaborating on the hypothesis that neurexins are essential for the coupling of synaptic vesicles to Ca<sup>2+</sup> channels, the authors show lowered Ca<sup>2+</sup> sensitivity in the TKO mice. Additionally, they reveal convincing evidence for the connection between the increased frequency of spontaneous IPSC and the higher number of glycinergic synapses of the LSO in the TKO mice, revealed by immunolabeling against the glycinergic presynaptic markers GlyT2 or VGAT.*

We thank the reviewer for the thoughtful and thorough evaluation of the significance of investigating the role of neurexins in glycinergic transmission at the MNTB-LSO synapse, particularly in the context of auditory processing and sound localization. The positive feedback is greatly appreciated.

**Weaknesses:**

*The major concern is novelty as this work on the effects of pan-neurexin deletion in a glycinergic synapse is quite consistent with the authors' prior work on glutamatergic synapses (Luo et al., 2020). The authors might want to further work out novel aspects and strengthen the comparative perspective. Conceptually, the authors might want to be more clear about interpreting the results on the altered dependence of release on voltage-gated Ca<sup>2+</sup> influx (Ca<sup>2+</sup> sensitivity, coupling).*

Regarding the reviewer's concern about the novelty of our work, we acknowledge that our previous work has explored the effects of pan-neurexin deletion on glutamatergic synapses (Luo et al., 2020). However, we would like to point out that a novelty of our present study indeed stems from the exploration of how different types of synapses converge to employ the same mechanism of synaptic function, particularly in the context of neurexin-mediated regulation. Our previous study focused on glutamatergic synapses, the current study delves into the realm of glycinergic synapses, which represent a distinct population with unique properties and functions. Despite the differences between these synapse types, our findings reveal a commonality in the underlying mechanisms of synaptic regulation mediated by neurexins. This convergence of mechanisms across different synapse types highlights the fundamental role of neurexins in synaptic function and plasticity. By elucidating how neurexins regulate synaptic transmission at both excitatory and inhibitory synapses, we provide valuable insights into the general principles governing synaptic function. In addition, this comparative perspective may shed light on the complex interplay between excitatory and inhibitory neurotransmission, which is crucial for maintaining the balance of neuronal activity and network dynamics.

**Recommendations for the authors:**

**Reviewer #1 (Recommendations For The Authors):**

*During the developmental period spanning P3-P12, the MNTB-LSO synapses undergo a transition from GABAergic to glycinergic transmission. It is well-established that Neurexin plays a role in modulating GABAergic transmission. In the authors' experimental system, AAV was injected at P0, likely impacting GABAergic transmission, including potentially influencing synapse number, before subsequently affecting glycinergic transmission. A*

*thoughtful discussion of how the experimental interventions might have influenced this developmental process and glycinergic transmission would enhance the clarity and interpretation of their findings.*

We thank the reviewer for raising the interesting topic of the transmitter switch during neurodevelopment. Strong evidence using gerbils and rats as animal models demonstrates that the MNTB-LSO synapses undergo a shift from GABAergic to glycinergic during the early development. However, in a more recent study by Friauf and colleagues (Fisher et al., 2019), patch-clamp recordings in acute mouse brainstem slices at P4-P11 combined with pharmacological blockade of GABAA receptors and/or glycine receptors clearly demonstrated no GABAergic synaptic component on LSO principal neurons, suggesting the transmitter subtype switch may be species different. We add a discussion in our revision to clarify this topic.

**Reviewer #2 (Recommendations For The Authors):**

*The data are compelling and report an intriguing functional phenotype. Mechanistic insight into how this phenotype manifests would significantly strengthen this study. For example, which neuroligin is found at these MNTB-LSO synapses?*

We agree that investigating the underlying molecular mechanisms, particularly the specific function of each variant of neuroligins and their respective ligands on the postsynaptic neurons, is crucial. Exploring these mechanisms, which extend beyond the scope of our current study, would undoubtedly enhance our understanding of neuroligin function at various synapses and foster advancements in the field.

*Does the TKO alter the ability of MNTB inputs to induce AP firing in LSO neurons?*

Activation of the MNTB inputs does not directly induce AP firing in LSO neurons, because the MNTB-LSO synapses are glycinergic and serve to inhibit neuronal activity.

We think the reviewer was to ask whether pan-neuroligin deletion in the MNTB neurons alter their ability to impact the firing of LSO neurons. Indeed, the weakening of glycinergic transmission due to pan-neuroligin ablation in MNTB neurons could potentially alter the excitation-inhibition (E/I) balance, thereby impacting the overall excitability of LSO neurons. We have conducted preliminary experiments to investigate this aspect and found that the E/I balance at LSO neurons was notably increased in TKO mice. We are currently preparing a manuscript to comprehensively address the role of neuroligins at the auditory circuit and behavior levels.

*Additional calcium measurements using GECIs would provide insight into whether nanodomain calcium or total calcium is altered at these synapses.*

We appreciate the valuable suggestion provided by the reviewer. However, distinguishing between  $\text{Ca}^{2+}$  nanodomain and  $\text{Ca}^{2+}$  microdomain using  $\text{Ca}^{2+}$  imaging techniques requires advanced systems such as two-photon STED microscopy, which are beyond the scope of our current research.

*It is unclear why fluorescence intensity is quantified instead of the number of synaptic clusters in LSO. In addition to changes in synapse numbers, fluorescent intensity can indicate a number of other possible morphological changes.*

We appreciate the valuable suggestion from the reviewer. We have re-analyzed our imaging data to compare synaptic density. The results, as included in Fig.3f and 3h, confirm an increase in the number of glycinergic synapses after pan-neuroligin deletion.

*The most robust synaptic phenotypes were produced by measuring light-evoked oIPSCs and the authors acknowledge that electrically-evoked eIPSCs might be contaminated by uninfected fibers or by other sources of glycinergic inputs. I suggest that IPSC PPRs, EGTA, and low Ca<sup>2+</sup> experiments be performed using optogenetics.*

As discussed in our response to Public Reviews, we acknowledge that optogenetic stimulation offers crucial advantages, and we have provided a balanced discussion of the caveats associated with both methods in our manuscript. Additionally, following the reviewer's suggestion, we have conducted new optogenetic experiments specifically for measuring the paired-pulse ratio in control and Nr<sub>1</sub>23 TKO mice. We included this new dataset in supplementary Figure S2, which is consistent with our result obtained with electrically fiber stimulation.

For experiments involving EGTA and low Ca<sup>2+</sup> manipulations, we opted for electrical stimulation due to major concerns regarding potential side effects of optogenetics, including the phototoxicity and photobleaching during prolonged light exposure.

*It is sometimes confusing which type of evoked stimulation is being used (e.g. PPR, EGTA, and low Ca<sup>2+</sup> experiments). To aid in the interpretations of these experiments, it would help to clarify.*

We appreciate the reviewer's suggestion regarding the clarity of the evoked stimulation methods used in our experiments. We have revised the manuscript to provide clearer descriptions of the specific types of evoked stimulation employed in each experiment. Thank you for guiding towards this clarification.

*The comparisons to Chen et al 2017 and the senior author's 2020 paper seem disjointed and do not contribute to the findings, which alone, are quite interesting. Given the prevailing notion that neurexins control different synaptic properties depending on the brain region and/or synapse studied, is it surprising that the findings observed here differ from previous studies of different synapses (glutamatergic and GABAergic)?*

By comparing previous studies at different types of neurons/synapses, our findings reveal a commonality in the underlying mechanisms of synaptic regulation mediated by neurexins. This convergence of mechanisms across different synapse types highlights the fundamental role of neurexins in synaptic function and plasticity. In addition, this comparative perspective may shed light on the complex interplay between excitatory and inhibitory neurotransmission, which is crucial for maintaining the balance of neuronal activity and network dynamics.

*Despite Nr<sub>1</sub>3 being the most abundant Nr<sub>1</sub> mRNA in MNTB neurons, the possible contributions of this highly expressed protein are not discussed.*

We thank the reviewer for bringing up this important point. The differential expression of individual neurexins indeed suggests that specific neurexins may play dominant roles in regulating synaptic transmission. While our study primarily focused on the collective impact of ablating all neurexins, we acknowledge the significance of exploring the specific contributions of individual neurexin isoforms in the future. Understanding the distinct roles of each neurexin isoform could provide valuable insights into the precise mechanisms underlying synaptic function and plasticity. We have added discussion in our revised manuscript Line223-230.

**Reviewer #3 (Recommendations For The Authors):**

- There are several instances of spaces missing and typos, please carefully check the manuscript.

We greatly appreciate the reviewer's helpful feedback on the text that could be clarified or improved. We have meticulously edited the manuscript to address these concerns.

- While studying the properties of IPSC, apart from optogenetic stimulation, the authors performed experiments with electrical fiber stimulation. Their findings showed a slightly significant reduction of the IPSC amplitude and no effect on the IPSCs kinetics when comparing the TKO and control. One weakness is the discrepancy between the results from the optogenetic and fiber stimulation experiments, which the authors contribute to inefficient transfection in the fiber stimulation experiments. The authors state that they tried to optimize their protocols for virus injection protocols. However, they do not elaborate on how the transfection rates could be improved in the discussion section. Moreover, it would be good to further address the reasons for the difference in amplitude between the control IPSCs in the optogenetic and fiber stimulation experiments.

Echoing the suggestion by Reviewer 2 (see above), we acknowledge that optogenetic stimulation offers certain advantages, and we have provided a balanced discussion of the caveats associated with both methods in our manuscript. In addition, we have performed a new set of optogenetic experiment for the paired-pulse ratio measurement in control and Nr1h3 TKO mice and included as a new figure in supplementary figure S2.

For experiments involving EGTA and low  $Ca^{2+}$  manipulations, we opted for electrical stimulation due to major concerns regarding potential side effects of optogenetics, including the phototoxicity and photobleaching during prolonged light exposure.

We added the detail of virus injection strategy that optimized the transfection rates in the method section "To enhance virus infection efficiency, we decreased the dosage per injection while increasing the frequency of injections. Additionally, we ensured the pipette remained immobilized for 20-30 seconds to guarantee virus absorption at injection sites. As a result of this strategy, we estimated that the vast majority of MNTB neurons were inoculated by AAVs." See line 288-290.

- Abstract: "ablation of all neurexins in MNTB neurons reduced not only the amplitude but also altered the kinetics of the glycinergic synaptic transmission at LSO neurons."

Changed as suggested.

- Consider revising to "The synaptic dysfunctions primarily resulted from an altered dependence of release on voltage-gated  $Ca^{2+}$  influx."

We appreciate the reviewer's suggestion, which helps improve the clarity of our manuscript. We have revised the phrasing as follows: "The synaptic dysfunctions primarily resulted from an impaired calcium sensitivity of release and a loosened coupling between voltage-gated calcium channels and synaptic vesicles."

- Line 39 should be vertebrates.

Revised as suggested.



- Line 49 it would sound better to say "which further points to the diverse actions of neurexins in specific neurons."

Revised as suggested.

- Line 60 - this paragraph could include information about GABA signaling from the MNTB to the LSO, because on line 113 you mention LSO neurons receive inhibitory GABAergic/glycinergic inputs, but when you do not mention blocking of GABA currents to isolate the glycinergic ones.

We thank the reviewer for the thoughtful and detailed suggestion. We revised the text in line 60 to "In the mature mammalian auditory brainstem" and in line 113, we removed GABAergic to emphasize the nature of glycinergic synapse, particularly in the mouse brainstem where no GABAergic components are found (Fisher et al., 2019).

- Line 72/73 it should be adeno-associated virus; line 73: "combining this with the RNAScope technique" sounds better.

Changed as suggested.

- Line 91 using the RNAScope technique; lines 97, 119 as a control; line 108 the functional organization.

Changed as suggested.

- Line 113 should be a pharmacological approach; line 122 optogenetically evoked.

Changed as suggested.

- Line 132, 160: the control.

Changed as suggested.

- Line 147 thus were infected; line 148 likely to be present but were obscured .

Changed as suggested.

- Line 154 which has been routinely used.

Changed as suggested.

- Line 155 It is not supposed to be Figure 2h but 2i; following that Figure 2i should be 2j; in my opinion, Figure 2i does not display a strong depression for the TKO mice.

Changed as suggested.

- Line 171 a better flow is achieved by saying: together these data show.

Changed as suggested.

- EC50 rather than IC50 of [Ca<sup>2+</sup>].

Changed as suggested.

| • *180 it is better to say "we approached the matter by..."; line 183 while recording;*

Changed as suggested.

| • *Line 203 were much stronger than the effect at control synapses; line 206 tightly clustering.*

Changed as suggested.

| • *Line 212 sounds like they provide evidence for retina and spinal cord as well, should be made clear.*

Changed as suggested.

| • *Line 289 previously.*

Changed as suggested.

| • *Line 295 should be 30 min.*

Changed as suggested.

| • *Line 336, 337 confocal microscope.*

Changed as suggested.

| • *Please provide the number of data points also in figure captions or in the results section.*

Added in the captions as suggested.

| • *Line 533, a better phrasing would be: the blocking effect of 0.2 mM Ca on IPSC amplitude.*

Changed as suggested.

| • *Explain either in the methods or result section how was the EC50 of Ca<sup>2+</sup> calculated.*

Added in the methods as suggested.